

Week Summary

Manager-worker and task-based parallelism

Summary

- This week we have seen alternatives to parallelising loops and domain decomposition
- OpenMP tasks allow algorithms and constructs to be parallelised which would otherwise be difficult or impossible
- Manager-worker and task-based parallelism offer more flexibility in how work is distributed which can improve performance
- Tasking used in the SWIFT cosmology code to achieve excellent strong scaling (See Schaller et al <https://dl.acm.org/doi/10.1145/2929908.2929916>)